



# Cambridge International AS Level

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## **MATHEMATICS**

**9709/22**

Paper 2 Pure Mathematics 2

**February/March 2026**

**MARK SCHEME**

Maximum Mark: 50

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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This document consists of **16** printed pages.

**PUBLISHED****Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

**GENERIC MARKING PRINCIPLE 1:**

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

**GENERIC MARKING PRINCIPLE 2:**

Marks awarded are always **whole marks** (not half marks, or other fractions).

**GENERIC MARKING PRINCIPLE 3:**

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

**GENERIC MARKING PRINCIPLE 4:**

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

**GENERIC MARKING PRINCIPLE 6:**

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

**Mathematics-Specific Marking Principles**

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

**Annotations guidance for centres**

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

**Annotations**

| <b>Annotation</b>                                                                   | <b>Meaning</b>                         |
|-------------------------------------------------------------------------------------|----------------------------------------|
|    | More information required              |
|    | Accuracy mark awarded zero             |
|    | Accuracy mark awarded one              |
|    | Independent accuracy mark awarded zero |
|  | Independent accuracy mark awarded one  |
|  | Independent accuracy mark awarded two  |
|  | Benefit of the doubt                   |
|  | Blank Page                             |
|  | Incorrect                              |
| Dep                                                                                 | Used to indicate DM0 or DM1            |

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| <b>Annotation</b>                                                                   | <b>Meaning</b>                                                                                                                             |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| DM1                                                                                 | Dependent on the previous M1 mark(s)                                                                                                       |
|    | Follow through                                                                                                                             |
|    | Indicate working that is right or wrong                                                                                                    |
| Highlighter                                                                         | Highlight a key point in the working                                                                                                       |
|    | Ignore subsequent work                                                                                                                     |
|    | Judgement                                                                                                                                  |
|    | Judgement                                                                                                                                  |
|    | Method mark awarded zero                                                                                                                   |
|    | Method mark awarded one                                                                                                                    |
|    | Method mark awarded two                                                                                                                    |
|    | Misread                                                                                                                                    |
|    | Omission or Other solution                                                                                                                 |
| Off-page comment                                                                    | Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to. |
| On-page comment                                                                     | Allows comments to be entered in speech bubbles on the candidate response.                                                                 |
|  | Judgment made by the PE                                                                                                                    |
|  | Premature approximation                                                                                                                    |
|  | Special case                                                                                                                               |
|  | Indicates that work/page has been seen                                                                                                     |

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| Annotation                                                                        | Meaning                                |
|-----------------------------------------------------------------------------------|----------------------------------------|
|  | Error in number of significant figures |
|  | Correct                                |
|  | Transcription error                    |
|  | Correct answer from incorrect working  |

**PUBLISHED****Mark Scheme Notes**

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

**Types of mark**

- M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
- B** Mark for a correct result or statement independent of method marks.
- DM or DB** When a part of a question has two or more ‘method’ steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- FT** Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.
- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
  - For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
  - The total number of marks available for each question is shown at the bottom of the Marks column.
  - Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
  - Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

**Abbreviations**

|        |                                                                                                                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AEF/OE | Any Equivalent Form (of answer is equally acceptable) / Or Equivalent                                                                                                                 |
| AG     | Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)                                                   |
| CAO    | Correct Answer Only (emphasising that no ‘follow through’ from a previous error is allowed)                                                                                           |
| CWO    | Correct Working Only                                                                                                                                                                  |
| ISW    | Ignore Subsequent Working                                                                                                                                                             |
| SOI    | Seen Or Implied                                                                                                                                                                       |
| SC     | Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance) |
| WWW    | Without Wrong Working                                                                                                                                                                 |
| AWRT   | Answer Which Rounds To                                                                                                                                                                |



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| Question | Answer                                                                                                  | Marks     | Guidance                             |
|----------|---------------------------------------------------------------------------------------------------------|-----------|--------------------------------------|
| 1        | Use a correct identity to obtain an equation involving $\tan \theta$ and $k$ only                       | <b>M1</b> | Allow presence of $\tan 135^\circ$ . |
|          | Obtain $\frac{\tan \theta - 1}{\tan \theta + 1} = k$ or equivalent involving only $k$ and $\tan \theta$ | <b>A1</b> |                                      |
|          | Obtain $\tan \theta = \frac{1+k}{1-k}$                                                                  | <b>A1</b> |                                      |
|          |                                                                                                         | <b>3</b>  |                                      |

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| Question | Answer                                                                                                | Marks     | Guidance                                                                    |
|----------|-------------------------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------------------|
| 2(a)     | V-shaped graph with vertex on positive $x$ -axis                                                      | <b>B1</b> |                                                                             |
|          | Draw correct graph of $y = 7 - 10x$                                                                   | <b>B1</b> | Correctly placed with respect to modulus graph and with a steeper gradient. |
|          |                                                                                                       | <b>2</b>  |                                                                             |
| 2(b)     | Attempt solution of linear equation with signs of $3x$ and $10x$ the same                             | <b>M1</b> |                                                                             |
|          | Obtain $-3x + 4 = 7 - 10x$ and hence $x = \frac{3}{7}$                                                | <b>A1</b> |                                                                             |
|          |                                                                                                       | <b>2</b>  |                                                                             |
| 2(c)     | State $3^{2t}$ equal to <i>their</i> answer from <b>2(b)</b> and apply logarithms to attempt solution | <b>M1</b> |                                                                             |
|          | Obtain $2t \ln 3 = \ln \frac{3}{7}$ or equivalent, and hence $t = -0.386$                             | <b>A1</b> | Allow greater accuracy $-0.38562...$                                        |
|          |                                                                                                       | <b>2</b>  |                                                                             |

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| Question | Answer                                                                                                                      | Marks      | Guidance                |
|----------|-----------------------------------------------------------------------------------------------------------------------------|------------|-------------------------|
| 3        | Apply logarithms to both sides to produce equation linear in $\sqrt{x}$ and $\ln y$                                         | <b>*M1</b> |                         |
|          | Obtain $\ln y = \ln k + a\sqrt{x}$                                                                                          | <b>A1</b>  |                         |
|          | Substitute values to obtain two simultaneous equations $2.83 = \ln k + 1.5a$ and $4.87 = \ln k + 3.2a$                      | <b>A1</b>  |                         |
|          | Attempt solution of simultaneous equations to find at least $a$ or $k$                                                      | <b>DM1</b> |                         |
|          | Obtain $a = 1.2$ and $k = 2.8$                                                                                              | <b>A1</b>  |                         |
|          | <b>Alternative Method for Question 3</b>                                                                                    |            |                         |
|          | Apply logarithms to both sides to produce equation linear in $\sqrt{x}$ and $\ln y$                                         | <b>*M1</b> |                         |
|          | Obtain $\ln y = \ln k + a\sqrt{x}$                                                                                          | <b>A1</b>  |                         |
|          | Find the gradient of the line and state $a = 1.2$                                                                           | <b>B1</b>  | Allow greater accuracy. |
|          | Substitute <i>their</i> value of $a$ into either $2.83 = \ln k + 1.5a$ or $4.87 = \ln k + 3.2a$ to obtain a value for $k$ . | <b>DM1</b> |                         |
|          | Obtain $k = 2.8$                                                                                                            | <b>A1</b>  |                         |
|          |                                                                                                                             | <b>5</b>   |                         |

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| Question | Answer                                                                                                                                                                | Marks      | Guidance                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------|
| 4        | Attempt use of product rule to differentiate $(x^2 - 3)\cos y$                                                                                                        | <b>*M1</b> | Allow a sign error but must have implicit differentiation. |
|          | Obtain $2x\cos y - (x^2 - 3)\sin y \frac{dy}{dx}$                                                                                                                     | <b>A1</b>  | OE                                                         |
|          | Differentiate $4\cos^3 y$ to obtain the form $k\cos^2 y \sin y \frac{dy}{dx}$                                                                                         | <b>M1</b>  | OE                                                         |
|          | Obtain complete $-12\cos^2 y \sin y \frac{dy}{dx}$ and 0                                                                                                              | <b>A1</b>  | OE<br>Allow unsimplified.                                  |
|          | Substitute $x = 2$ and $y = \frac{1}{3}\pi$ to find gradient of curve                                                                                                 | <b>DM1</b> |                                                            |
|          | Obtain $2 - \frac{1}{2}\sqrt{3} \frac{dy}{dx} - \frac{3}{2}\sqrt{3} \frac{dy}{dx} = 0$ and hence $\frac{1}{\sqrt{3}}$ , $\frac{\sqrt{3}}{3}$ or $\frac{1}{3}\sqrt{3}$ | <b>A1</b>  |                                                            |
|          |                                                                                                                                                                       | <b>6</b>   |                                                            |

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| Question | Answer                                                                                                        | Marks     | Guidance                                          |
|----------|---------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------|
| 5(a)     | Substitute $x = -\frac{1}{2}$ , equate to $-3$ and attempt solution                                           | <b>M1</b> |                                                   |
|          | Obtain $-\frac{1}{8}a + \frac{1}{4} - \frac{1}{2}(a+1) + 1 = -3$ and hence $a = 6$                            | <b>A1</b> |                                                   |
|          |                                                                                                               | <b>2</b>  |                                                   |
| 5(b)     | Divide <i>their</i> $p(x) + 3$ by $2x + 1$ at least as far as the $x$ term in the quotient                    | <b>M1</b> | Or equivalent method, e.g. inspection.            |
|          | Obtain $3x^2 - x + 4$ and conclude $(2x + 1)(3x^2 - x + 4)$                                                   | <b>A1</b> |                                                   |
|          |                                                                                                               | <b>2</b>  |                                                   |
| 5(c)     | Equate $\frac{1}{2 + \operatorname{cosec} \theta}$ to $-\frac{1}{2}$ only and attempt value for $\sin \theta$ | <b>M1</b> | Must be using a correct method of simplification. |
|          | Obtain $\sin \theta = -\frac{1}{4}$ and hence $\theta = 194.5$                                                | <b>A1</b> | Allow greater accuracy 194.4775...                |
|          |                                                                                                               | <b>2</b>  |                                                   |

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| Question | Answer                                                                                                                                                          | Marks      | Guidance |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|
| 6        | Attempt to express integrand in terms of $\cos 4x$ and $\sec^2 x$                                                                                               | <b>M1</b>  |          |
|          | Obtain $2\cos 4x + 3\sec^2 x - 1$                                                                                                                               | <b>A1</b>  | OE       |
|          | Integrate to obtain the form $k_1 \sin 4x + k_2 \tan x + k_3 x$                                                                                                 | <b>*M1</b> |          |
|          | Obtain correct $\frac{1}{2} \sin 4x + 3 \tan x - x$                                                                                                             | <b>A1</b>  |          |
|          | Apply limits correctly to find an exact value                                                                                                                   | <b>DM1</b> |          |
|          | Obtain $-\frac{1}{4}\sqrt{3} + 3\sqrt{3} - \frac{1}{3}\pi - (\frac{1}{4}\sqrt{3} + \sqrt{3} - \frac{1}{6}\pi)$ and hence $\frac{3}{2}\sqrt{3} - \frac{1}{6}\pi$ | <b>A1</b>  | ISW      |
|          |                                                                                                                                                                 | <b>6</b>   |          |

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| Question | Answer                                                                                                                       | Marks      | Guidance                                          |
|----------|------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------|
| 7(a)     | Integrate to obtain the form $k_1e^{2x} + k_2e^x$                                                                            | <b>*M1</b> |                                                   |
|          | Obtain correct $\frac{1}{4}e^{2x} + \frac{1}{4}e^x$                                                                          | <b>A1</b>  |                                                   |
|          | Apply limits correctly and equate to 50                                                                                      | <b>DM1</b> |                                                   |
|          | Obtain $\frac{1}{4}e^{4a} - \frac{1}{4}e^a = 50$ or similarly simplified equivalent                                          | <b>A1</b>  |                                                   |
|          | Rearrange to confirm $a = \frac{1}{4}\ln(200 + e^a)$                                                                         | <b>A1</b>  | <b>AG</b> – necessary detail needed.              |
|          |                                                                                                                              | <b>5</b>   |                                                   |
| 7(b)     | Consider sign of $a - \frac{1}{4}\ln(200 + e^a)$ , or equivalent for values 1 and 1.5                                        | <b>M1</b>  |                                                   |
|          | Obtain $-0.32\dots$ and $0.16\dots$ or equivalents, and justify conclusion                                                   | <b>A1</b>  |                                                   |
|          |                                                                                                                              | <b>2</b>   |                                                   |
| 7(c)     | Use iteration process correctly at least once                                                                                | <b>M1</b>  |                                                   |
|          | Obtain final answer 1.329                                                                                                    | <b>A1</b>  | Answer required to exactly 4 significant figures. |
|          | Show sufficient iterations to 6 significant figures to justify answer, or show sign changes in the interval [1.3285, 1.3295] | <b>A1</b>  |                                                   |
|          |                                                                                                                              | <b>3</b>   |                                                   |

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| Question | Answer                                                                                                                       | Marks      | Guidance                                                |
|----------|------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------|
| 8        | Equate $y$ to 4 and attempt solution to find at least the value of $\ln x$                                                   | <b>M1</b>  |                                                         |
|          | Obtain $\ln x = 3$ and $x = e^3$                                                                                             | <b>A1</b>  | Or exact equivalent.                                    |
|          | Use correct quotient rule, or equivalent, to attempt expression for first derivative                                         | <b>*M1</b> |                                                         |
|          | Obtain correct $\frac{-\frac{8}{x}}{(1 + \ln x)^2}$ or (unsimplified) equivalent                                             | <b>A1</b>  |                                                         |
|          | Substitute their $x = e^3$ and $\ln x = 3$ to find gradient                                                                  | <b>DM1</b> |                                                         |
|          | Obtain gradient $-\frac{1}{2}e^{-3}$ or exact equivalent                                                                     | <b>A1</b>  |                                                         |
|          | Attempt equation of tangent and substitute $y = 0$ to find $x$ -value                                                        | <b>M1</b>  | Or using a right-angled triangle and gradient of $PQ$ . |
|          | Obtain $-4 = -\frac{1}{2}e^{-3}(x - e^3)$ or $\frac{4}{e^3 - q} = -\frac{1}{2}e^{-3}$ or equivalent, and hence answer $9e^3$ | <b>A1</b>  |                                                         |
|          |                                                                                                                              | <b>8</b>   |                                                         |